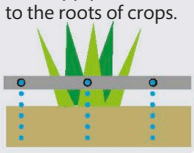
Irrigation

Some farmers rely on natural rainfall to water their crops, but in some climates, crops may also require irrigation systems. These vary from simple gravity-fed methods to those that apply water directly to the roots of each plant. Irrigation can be problematic: water can be wasted, crops can be contaminated if untreated waste water is used, and salinity can build up in the soil. Smart technology can be used to deliver water where it is needed most, rather than providing blanket coverage.

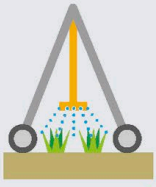
Surface
Water floods the entire surface or runs down furrows through gravity or pumping. This is highly labor-intensive, much water is lost to evaporation and runoff, and there is a risk of waterlogging.



Drip
A drip system uses pipes, either made of a porous material or perforated, which are placed on or under the ground to apply water directly to the roots of crops.

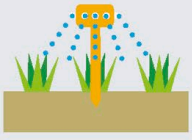


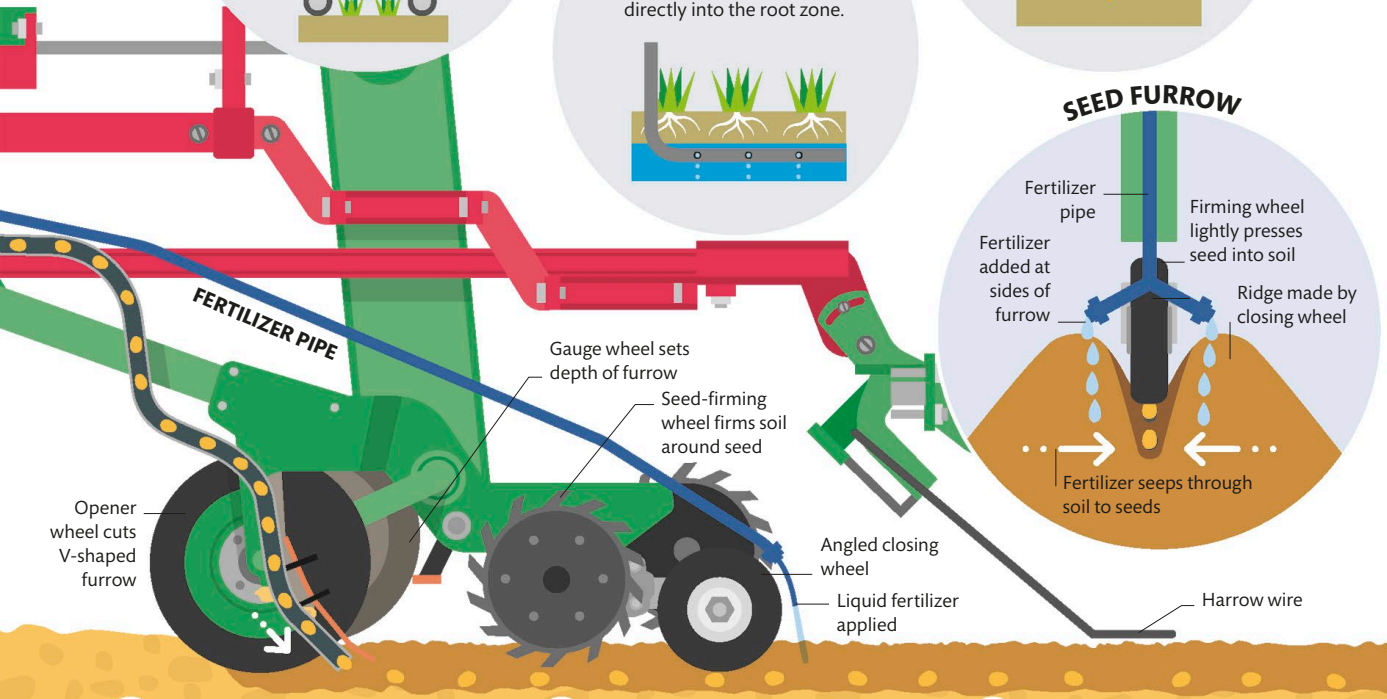
Center pivot
Sprinklers move on wheeled towers in a circular motion. This method can water a large area in a relatively short time.



Subirrigation
An underground porous-pipe system is used to raise the water table or to discharge water directly into the root zone.

Sprinkler
Water is distributed by high-pressure overhead sprinklers or guns on moving platforms. However, water can be lost when sprayed up in the air.





5 Cutting the furrow
Furrows are cut using wheels or blades to open the soil to the right depth and shape. Seeds are dropped at regular intervals behind the opener wheel. Fertilizer and insecticide are sometimes added.

6 Firming the seeds
A seed-firming wheel presses the seed into the furrow using a rolling or sliding action to improve its contact with the soil and moisture at the bottom of the furrow. It also stops the seed from bouncing out.

7 Closing and fertilizer delivery
Angled closing wheels press soil firmly around the seed. If fertilizer is not applied with the seed, it is now added to one or both sides of the furrow. The surface is then leveled by a roller or harrow wire.